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Purification of proteins using peptide-ELP based affinity precipitation

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ABSTRACT

In this work, we employed fusions of affinity peptides and elastin-like polypeptide (ELP) to carry out a proof-of-concept study for the single-step purification of model, tag-free proteins. Three known peptide-protein binding pairs of varied binding strengths were evaluated. The peptide-protein binding was first characterized through in-solution binding techniques such as fluorescence spectroscopy. Peptide-ELP constructs were then produced in *E. coli* and purified by phase transition. The binding of the peptide-ELP constructs to the products were then evaluated using competitive fluorescence spectroscopy. Affinity capture, precipitation and recovery experiments were then conducted using the three constructs to evaluate the efficacy of these peptide-ELP based affinity precipitation processes. Two out of the three systems tested were successful in capturing the product and yielding pure protein in a single affinity precipitation step. These results indicated that peptide-protein affinity played an important role in both the effective capture of the protein, and also in the required binding molar ratio for the affinity precipitation process. In addition, for intermediate affinity systems, constructs containing two copies of the peptide at the N-terminus of the ELP were beneficial towards achieving both high purity and yield, likely due to increased affinity from avidity effects. This proof-of-concept study lays the foundations for the development of new peptide-ELP-based affinity purification processes for industrially relevant proteins and other classes of biologics.

1. Introduction

Elastin-like polypeptides (ELP) are a class of "smart" biopolymers, which possess the ability to undergo phase transition from soluble to precipitate form when the temperature is elevated above a transition temperature. The sequence of ELP is characterized by the repeating unit V-P-G-X-G, where the "guest residue" X can be modified to any amino acid (except proline) to shift the transition temperature (T_t) (Urry, 2011; Meyer and Chilkoti, 2004). The 'salting-out' effect (Arakawa and Timasheff, 1984) can also be employed to induce phase transitions with ELPs at relatively low ionic strengths (e.g. 0.2 M Na_2SO_4) (Sheth et al., 2013). The temperature/salt-induced phase-transition of ELP has been shown to be reversible as it resolubilizes when the temperature is reduced below the transition temperature or when the salt concentration is decreased (Meyer and Chilkoti, 1999).

Studies over the past two decades have demonstrated a variety of ways to purify proteins using ELPs. Meyer and Chilkoti employed ELP fusions with the protein to enable direct precipitation of the product from complex mixtures (Meyer and Chilkoti, 1999). In their work, proteases were required to cleave the ELP from the desired protein product. Wood and co-workers have used a combination of ELP and self-cleaving inteins or split inteins for the purification of proteins (Coolbaugh et al., 2017; Shi et al., 2013). Their approach obviates the need for protease treatment. Kim and Chen have developed an affinity capture-based purification platform using peptide-tagged proteins in concert with ELPs fused with complementary tag-binding SH3 domains (Kim and Chen, 2016). In this approach the purified product retains the peptide tag. An alternative approach to using product-ELP fusions is to employ affinity ligand-ELP constructs to capture and precipitate the desired product, eliminating the need for subsequent cleavage of the ELP from the product. This approach has been successfully employed for the purification of monoclonal antibodies (mAbs) using ELP coupled to the Z domain of protein A (Sheth et al., 2014b; Swartz et al., 2018a). Further, this construct has been shown to produce comparable purity of mAbs and Fc-fusion proteins as compared to traditional protein A chromatography and has been scaled using appropriate membrane systems (Sheth et al., 2014a; Swartz et al., 2018b).

While protein A can be employed for the purification of most mAbs,

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many proteins currently lack platform purification processes. The current work explores the use of alternative affinity ligands such as peptides for the purification of proteins by affinity precipitation using the phase-transition properties of ELPs. Affinity peptides can be readily identified using a variety of design approaches and high-throughput screening (HTS) formats (Chandra et al., 2013, Chandra et al., 2014). Peptides as affinity reagents have many desirable properties such as ease of synthesis, stability, and relatively low cost. Several researchers have developed affinity peptides for the purification of proteins. A cyclic disulfide-constrained peptide was identified and utilized for the chromatographic purification of recombinant Factor VIII from complex industrial feeds over many column cycles (Kelley et al., 2004). Solid-phase peptide synthesis (SPPS) has also been carried out directly on CIM monolith disks to develop affinity peptide matrices for the purification of tissue plasminogen activator (Vlakh et al., 2004). Carbonell and co-workers have developed affinity peptides for several proteins (IgG and erythropoietin) and have demonstrated their utility in column format for purification of these and other classes of biological products (Islam et al., 2019; Kish et al., 2018). We have recently employed rational design and novel screening protocols for the development of peptide ligands for the affinity chromatographic purification of biologics such as human growth hormone (Chandra et al., 2019) and Fab fragments (Nascimento et al., 2019).

The current work is the first demonstration of a peptide-ELP-based affinity precipitation process for protein purification. We employed a combination of ELPs and peptides to carry out a proof-of-concept study for the single-step purification of model, tag-free proteins. The ELP used in the current work is ELP-$(KV_8F)_{40}$, where KV_8F denotes the substitutions and frequencies of the position of the guest residue 'X' in V-P-G-X-G and '40′ is the number of V-P-G-X-G moieties in the ELP construct (Lim et al., 2007). Further, we have also studied the ability of peptide-ELP fusions to purify proteins from crude mixtures by affinity precipitation. Three known peptide-protein binding pairs – Flag peptide-Anti-Flag M2 (Hopp et al., 1988), DVEAWLDERVPLVET-streptavidin (Lamla and Erdmann, 2004) and SMWRTYI-human growth hormone (Chandra et al., 2019) – of varied binding strengths were chosen and evaluated. While the first two pairs were obtained from previous studies in the literature, the SMWRTYI peptide was developed previously in our lab using rational design strategies to purify hGH in a chromatographic format. The peptide-protein binding was first characterized through in-solution binding techniques such as fluorescence spectroscopy. Peptide ELP constructs are then produced in *E. coli* and purified by phase transition. The binding of the peptide ELP constructs to the products is evaluated using competitive fluorescence spectroscopy. Affinity capture, precipitation and recovery experiments were then performed using the three constructs to demonstrate the feasible operation of peptide-ELP-based affinity precipitation.

2. Materials and methods

2.1. Materials

Amicon® Ultra-15 centrifugal filter units (NMWL/MWCO of 30 and 100 kDa) were obtained from Merck Millipore (Darmstadt, Germany). Citric acid, tribasic sodium citrate dihydrate, Tris−HCl, Trizma base, sodium hydroxide, L-cysteine, sodium phosphate (monobasic and dibasic), sodium sulfate, triisopropyl silane (TIPS), trifluoroacetic acid (TFA), dichloromethane (DCM), and Anti-Flag antibody M2 were purchased from Sigma Aldrich (St. Louis, MO/USA). Recombinant streptavidin was obtained from G-Biosciences. Human growth hormone (hGH) was generously donated by Novo Nordisk. 10X SDS PAGE running buffer, Kaleidoscope Precision-Plus protein gel standard ladder, Bio-Rad AnykD polyacrylamide gel and Mini Protean Tetra-cell two gel system were procured from Bio-Rad Laboratories. Yeast extract was obtained from Amresco, while Bacto-tryptone, Bacto-agar were obtained from BD.

Single stranded oligonucleotide sequences and gBlock gene fragments encoding the peptide/linker sequences were purchased from Integrated DNA Technologies (IDT). Pfu Ultra II Hotstart PCR Master Mix for cloning was obtained from Agilent Technologies. DH5α variant NEB5α cells, DpnI, BamHI, EcoRI, NdeI and PstI restriction enzymes were purchased from New England Biolabs (Ipswich, MA). QiaPrep spin Miniprep kit was ordered from Qiagen (Frederick, MD). Bio-Rad Laboratories MyCycler was used for carrying out PCR. Solid black 384 well plates (CAT 3575) from Corning (Tewksbury, MA) were used for fluorescence polarization assays.

Peptide synthesis reagents: Dimethylformamide (DMF), piperidine, N-methyl pyrrolidone (NMP), activator 2-(1*H*-benzotriazol-1-yl)-1,1,3,3-tetramethyluronium hexafluorophosphate (HBTU) and methyl-tertiary-butyl ether (MTBE) were purchased from AGTC Bioproducts (Wilmington, MA). Nova-PEG Rink Amide Resin was obtained from EMD MilliporeSigma (Burlington, MA). All 20-fluorenylmethoxy carbonyl (Fmoc) amino acids were obtained from 21 st Century Biochemicals (Marlborough, MA). Fmoc-Lys-FAM−OH was purchased from AAT Bioquest (Sunnyvale, CA).

2.2. Solid-phase peptide synthesis (SPPS) and peptide purification

Peptides were synthesized with a C-terminal 5(6)-carboxyfluorescein-labeled lysine, by solid phase peptide synthesis (SPPS) on an Intavis AG Multipep RS (Chicago, IL). Fluorenylmethoxycarbonyl (Fmoc) chemistry was used to synthesize the peptides. The synthesis protocol was adapted from prior work performed in our lab (Chandra et al., 2013). Peptide purity was determined by a Waters Acquity ultra-performance liquid chromatography (UPLC) system on a Waters BEH C18 reversed phase liquid chromatography (RPLC) column. Impure peptides (less than 70 % purity) were purified on a Shimadzu HPLC system by reversed phase liquid chromatography using a Waters reversed phase BEH C18 column. Fractions were analyzed by RP-UPLC for purity and by MALDI-ToF on a Bruker MALDI-Tof II mass spectrometry system for identity confirmation.

2.3. Fluorescence polarization for determination of peptide-protein dissociation constants (K_d)

Peptide-protein affinities were determined by fluorescence polarization. 25–250 nM solutions of fluorescently labeled peptides were prepared. 20 μL of the peptide was then incubated with 20 μL of 12 serial dilutions (0.005 − 9 μM for the Flag system, 0.003 − 55 μM for the streptavidin system and 0.06 − 125 μM for the hGH system) of the protein in black, 384 well plates. A 4–5 log range of protein concentration was covered in the binding experiments. The peptide-protein mixtures were incubated with shaking, and fluorescence measurements were recorded after 3 h. Results were obtained in the form of either a change in fluorescence polarization, or in fluorescence intensity using a Biotek Synergy HT plate reader (city, state). Dissociation constants were then determined by fitting a 4-parameter logistic curve (Findlay and Dillard, 2007; Gadagkar and Call, 2015) to the data (equation 1). The 'C' parameter in the equation represents the dissociation constant K_d of the interaction. Appropriate controls such as dye-protein and protein-buffer interactions were also performed to ensure that the results obtained corresponded to specific peptide-ELP protein interactions.

$$Y = \frac{(A - D)}{\left(1 + \left(\frac{X}{C}\right)^{B}\right)} + D$$

Eq. 1: 4 parameter logistic curve equation used to fit binding curve data. A = response at zero analyte concentration, B = slope factor, C = inflection point, D = response at infinite analyte concentration

2.4. Cloning and expression of peptide-ELP constructs

The plasmid encoding the ELP-$(KV_8F)_{40}$ (Liu and Chen, 2013) was provided by our collaborators at the Chen laboratory at University of Delaware. Primers corresponding to the peptides were obtained from Integrated DNA Technologies. Pfu-Ultra Hotstart PCR Master Mix was used for the recombinant cloning of the peptides into the plasmid by insertion or replacement site-directed mutagenesis. A Glycine-Serine linker was also cloned in at the N-terminal end of the ELP and retained in all subsequent peptide-ELP fusions. Cloning of the SMWRTYI and DYKDDDDK peptides was performed in a single step of insertion site-directed mutagenesis. Due to a longer sequence, the streptavidin-binding peptide DVEAWLDERVPLVET (strep-pep) was cloned into the plasmid in two steps using two sets of primers. Bacterial transformation and sequencing were performed following the insertion reactions to confirm the identity of the peptide-ELP fusions. For the dipeptide-ELP fusion for the streptavidin system, a double stranded DNA insert was ligated into a vector linearized by NdeI and PstI restriction endonucleases.

Single colonies of BL21DE3 *E. coli* corresponding to the respective constructs were picked from 50 μg/mL Kanamycin LB agar plates and grown in 5 mL cultures of LB media. The cultures were grown overnight before being sub-cultured into 250 mL Terrific Broth (TB). They were then grown for 3 − 4 hours at 37 °C and induced with 1 mM IPTG at OD_{600} of 0.8–1.2 for 12 − 14 hours at 25 °C. Subsequent cell lysis and purification was carried out as mentioned by Sheth et al. (2014a,b) performing at least two cycles of inverse transition cycling (ITC). The expressed and purified constructs were analyzed by RP-UPLC on a C4 column to determine purity.

2.5. Competitive binding experiments for determination of peptide-ELP-protein binding affinity

Fluorescence spectroscopy, in concert with the BioEQS software (Royer, 1993), was used for the determination of binding affinity between the peptide-ELP fusion and the protein. Results from peptide-protein binding experiments (described above) guided the selection of appropriate starting concentration for the competitive binding assays. Typical fluorescence spectroscopy-based competitive experiments involved incubating the peptide and protein in a ratio such that 70–80 % of the available binding sites were occupied as indicated by the saturation of the fluorescence polarization/quenching signal. To this mixture containing majority of bound protein-peptide complexes, the competitor (in this case the peptide-ELP conjugate) was added until the fluorescence/polarization signal completely decayed to the baseline value. This indicated complete knocking off of the protein bound-labeled peptide by the peptide-ELP construct with the latter now occupying the binding sites on the protein.

15 μL of a low concentration of the labeled peptide was incubated with 15 μL of the corresponding protein at a concentration such that the fluorescence polarization value reached ~80 % of the maximum signal. Several such samples were prepared in a solid black flat-bottomed 384-well plate (Corning). A wide range of concentrations of the competing peptide-ELP construct was prepared, and 20 μL of each concentration was added to one well of the peptide-protein mixture. The decrease in polarization value across increasing competitor concentration was recorded. The BioEqs software (Royer, 1993) was employed for data fitting and parameter estimation and included the labeled Flag peptide, Antibody M2 and Flag-ELP species as well as bound complexes. The values were input into the software and the binding affinity of the peptide-ELP (competitor) and corresponding protein was determined.

2.6. Peptide-ELP binding, precipitation and purification experiments

All reactions were performed in 0.5- or 1.5-mL microcentrifuge tubes. Fig. 1 outlines a general schematic of the steps followed for peptide-ELP based affinity precipitation. The binding step involved incubating 50 μL of peptide-ELP (Flag-ELP, Str-ELP or SMWRTYI-ELP) with 50 μL of the corresponding protein solution in various molar ratios ranging from 1:1 to 40:1 of peptide-ELP to protein. Binding was carried out at room temperature in PBS pH 7.4 for 1 h. Anti-Flag Antibody M2 was used at a concentration of 5 μM, while streptavidin and hGH were typically used at a concentration of 20 μM (of tetrameric streptavidin). After binding, 50 μL of 1 M Na_2SO_4 in PBS pH 7.4 was added to the binding mixture to obtain a final concentration of 0.33 M Na_2SO_4 and to precipitate the complex. The slurry was subsequently centrifuged for 15 min at 10,000 g to pellet the precipitated complex. The binding supernatant (S1) was removed and the pelleted precipitate was re-solubilized in 100 μL of the respective elution buffers (pH 4.8–5.0 50 mM citrate for the streptavidin system and pH 3.0 50 mM citrate for the Flag and hGH systems). Following re-solubilization and dissociation of the complex, 50 μL of 1 M Na_2SO_4 at the elution conditions was added to the solution to precipitate the construct.

The slurry was then centrifuged for 15 min at 10,000 g to pellet the peptide-ELP construct, leaving the corresponding protein (streptavidin, Anti-Flag antibody M2 or hGH) in the supernatant. The supernatant was aspirated and the pelleted peptide-ELP precipitate was resolubilized in 100 μL PBS pH 7.4. Concentrations of all fractions were evaluated using C_4 RP-UPLC and protein recovery was determined.

For protein purification experiments carried out from complex mixtures, the proteins were spiked into *E. coli* cell lysate at the same concentrations as performed in pure component experiments (described in the paragraph above). The entire binding and precipitation procedures carried out were as described above, with the only difference being the inclusion of 3 wash steps with 0.33 M Na_2SO_4 in PBS pH 7.4 after the first precipitation step.

2.7. SDS PAGE analysis

Denaturing, non-reducing SDS polyacrylamide gel electrophoresis was performed using Bio-Rad AnyKd polyacrylamide gels on the Mini Protean Tetra-cell system. Samples were diluted 1:1 in a 2x Laemmli sample buffer and denatured at 95 °C for 10 min. 25 μL of the resulting samples were loaded into the wells and the gel was run in a 1X SDS Tris Glycine running buffer at room temperature for 30 min. Gel staining was performed using Coomassie blue and the gels were imaged in the BioRad gel documentation system.

2.8. Reversed-phase UPLC analysis

RP-UPLC analyses were carried out to determine the purities of the synthesized peptides as well as the bi-functional peptide-ELP constructs and protein purities and yields. For the synthesized peptides, a Waters BEH analytical C_{18} column was used with a 5-minute method containing a column equilibration step with 100 % buffer A (5 % acetonitrile in water with 0.1 % trifluoroacetic acid), a linear gradient from 100 % buffer A to 100 % buffer B (95 % acetonitrile in water with 0.1 % trifluoroacetic acid) and a column regeneration step with 100 % buffer B. For the ELP constructs and analysis of protein purity and yield, a C4 column was employed using a 15-minute gradient method comprised of the same steps as above.

3. Results and discussion

Although we have previously demonstrated that the phase transition biopolymer ELP can be employed in a fusion protein format with the Z domain of protein A for mAb purification, the use of smaller affinity ligands such as peptides in this format has not been demonstrated to date. This paper addresses two key questions: 1) can affinity peptides be employed in an ELP fusion format for protein purification and 2) the impact of their target binding on the efficacy of the affinity precipitation process. Three peptides were evaluated; a high affinity Flag peptide

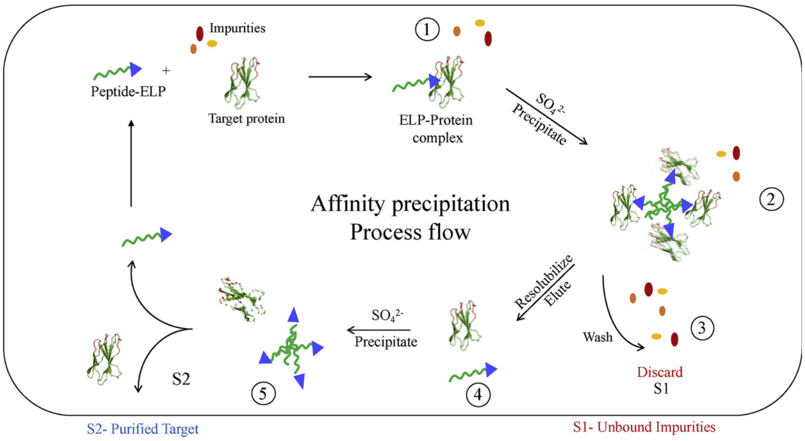


Fig. 1. Schematic for peptide-ELP based affinity precipitation involving 5 steps of 1) binding of peptide-ELP to protein of interest, 2) precipitation of bound complex, 3) washing away of impurities, 4) elution/dissociation of protein from peptide-ELP and 5) re-precipitation of the peptide-ELP construct. S1 is the impurity-containing unbound supernatant, which is discarded. S2 is the second supernatant, which contains the purified product. The colored ellipses represent impurities while the target protein is indicated by the cartoon representation of a protein.

(Hopp et al., 1988) for a specific mAb anti-Flag antibody M2 (note: the Flag peptide is known to bind to the CDR of the antibody M2 (Entzminger et al., 2017), a moderate affinity streptavidin-binding peptide (Lamla and Erdmann, 2004), and a relatively low affinity peptide that binds hGH (Chandra et al., 2019; Ko et al., 2017). First, binding affinities of these three peptides for their target proteins were determined. Next, ELP fusion proteins containing the peptides were expressed, purified, and characterized for their protein binding in solution using fluorescence spectroscopy. Finally, they were employed in precipitation and recovery experiments as indicated in Fig. 1 to evaluate their efficacy for protein purification.

3.1. Peptide-protein binding affinity measurements

Prior to expressing and evaluating the peptide-ELP reagents, it was important to evaluate the binding affinity of the three peptides to their respective protein targets. The binding affinities were evaluated by fluorescence spectroscopy as described in the methods section using peptides that were labeled at their C-termini with 5(6)-FAM. Fig. 2a shows the fluorescence polarization (FP) data for the binding of the Flag peptide to the Anti-Flag antibody M2 across a range of nine (0.005 – 9 μM) concentrations of the antibody. The four-parameter logistic equation given in the methods section (equation 1) was employed to fit the curve and a K_d value of 99.4 ± 12 nM was obtained. This affinity measurement is within the same order of magnitude as that reported by Tatsumi et al. via SPR (Tatsumi et al., 2017). Interestingly, for binding experiments carried out with streptavidin and its binding peptide (strep-pep), no clear shifts in polarization of FAM-labeled peptide were observed upon binding to streptavidin. This is not surprising given that the fluorescence properties of the same dye can vary across different solution types and binding conditions. However, a change in fluorescence intensity of the FAM dye with streptavidin concentration was seen which is not unusual, since fluorophores can undergo quenching depending on the nature of the binding site involved. As shown in Fig. 2b, the K_d value obtained from the free solution fluorescence quenching experiments for the binding of strep-pep to streptavidin was 7.54 ± 0.025 μM. This value is higher than that reported in the literature (Erdmann and Lamla, 2011), likely due to avidity effects and the presentation technique employed in the prior SPR studies where the immobilized peptide was observed to have stronger binding. Finally, the K_d value for the peptide-hGH system was also determined using FP (Fig. 2c) and was found to be 31.9 ± 0.12 μM. This affinity measurement was corroborated by previous experiments conducted in our lab for the design and application of the SMWRTYI peptide for hGH purification and analysis (Chandra et al., 2019; Ko et al., 2017). Overall, these results confirmed that the three selected peptides exhibited a range of binding affinities for their respective protein targets. ELP fusion proteins containing these peptides

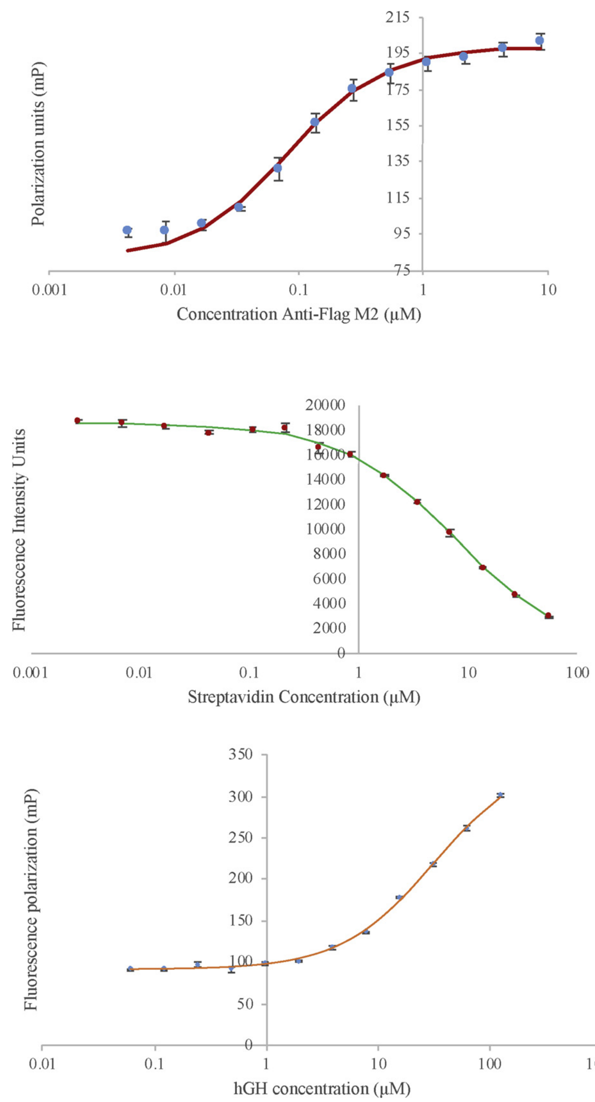


Fig. 2. Measurement of protein-peptide binding affinity via fluorescence polarization or quenching experiments conducted at pH 7.4 for all 3 peptide-protein pairs using fluorescently-labeled peptides. (a) Flag peptide (DYKDDDDK) binding to anti-Flag antibody M2 (n = 4), (b) Binding of DVEAWLDERVPLVET peptide to streptavidin (n = 2) and (c) Binding of SMWRTYI to human growth hormone (hGH) (n = 4).

Table 1
Peptide-protein binding dissociation constants for the different pairs obtained in this work compared to previous values from literature with relevant references. Peptide-ELP-protein affinities have also been indicated.

Protein Target	Peptide Sequence	Peptide-target K_d (from literature)	Technique used in literature	Peptide-target K_d by fluorescence (current work)	ELP-peptide-target K_d
Anti-Flag Antibody M2	DYKDDDDK (Flag peptide)	3–28 nM	SPR	99.4 ± 12 nM	~49.6 nM
Streptavidin	DVEAWLDERVPLVET	4 nM	SPR	7.54 ± 0.025 μM	41.6 μM[a]
hGH	SMWRTYI	27 μM	FP	31.9 ± 0.12 μM	ND

ND- Not determined.
[a] Estimated apparent K_d, due to multimeric nature of the product.

were then expressed and purified as described in the next section. Estimated dissociation constants for the peptide-protein and peptide-ELP-protein systems are indicated in Table 1. They are also compared with values obtained from the literature.

3.2. Expression and purification of peptide-ELP fusions

The protocol described in the methods section was employed to clone the peptides at the N-terminus of the ELP using site-directed, mutagenesis-based cloning. The N-terminal attachment was carried out to facilitate exposure of the peptide's N-terminus and ensure effective presentation of the ligand to the target in solution. Amino acid spacers of different lengths were incorporated between the peptide and ELP to impart flexibility to the ligands and maintain sufficient distance from the hydrophobic ELP. Flexible glycine-serine linkers were used in the case of the Flag-ELP (GS6) and the strep-pep-ELP systems (GS6 and GS10) while a variety of linkers (flexible Gly-Ser linkers, rigid Pro-Ala linkers, and charged Lys or Asp linkers) were tested in the SMWRTYI-ELP system. The resulting peptide-ELP fusions were expressed and purified as described in the methods Section 2.4. Expression yields greater than 100 mg/L of shake flask bacterial culture were observed for all three fusions. Importantly, RP-UPLC analyses and MALDI-ToF mass spectrometric analysis of the constructs indicated high purity of the peptide-ELP fusions after several rounds of inverse transition cycling purification (SI figures).

3.3. Binding affinity of peptide-ELP to protein

The attachment of the peptide to the ELP can potentially impact the binding affinity to the target protein. Steric hindrances, non-specific binding by the ELP, or inadequate presentation of the affinity peptide can result in decreased affinity. On the other hand, additional stabilizing interactions from the ELP residues near the peptide may increase the binding affinity due to entropic effects. The binding affinity of the peptide-ELP fusions for their respective proteins was evaluated using competitive FP experiments. Various concentrations of peptide-ELP were added to pre-incubated mixtures of labeled peptides and proteins. Control experiments were carried out to determine dye-protein and protein-buffer interactions and no non-specific interactions were observed. Fig. 3a presents the binding results, obtained from FP, for the Flag-ELP fusion experiments. As can be seen, addition of Flag-ELP resulted in a decrease in polarization, approaching the value for the free, labeled peptide at higher Flag-ELP concentrations. This indicates complete displacement of the labeled, Flag peptide by Flag-ELP. Interestingly, the apparent K_d value (of Flag-ELP for M2 Anti-Flag mAb) obtained from this analysis was 49.6 nM which was stronger than that obtained for the binding of the labeled Flag peptide to the antibody (99.4 nM). This improved, apparent affinity suggests that the attachment of the peptide to the ELP potentially enhanced the binding of the peptide to the antibody. The Flag peptide (DYKDDDDK) is highly hydrophilic, and the ELP moiety (V-P-G-X-G) considerably hydrophobic. The enhanced affinity of the Flag-ELP as compared to the peptide alone could be due to additional interactions of the ELP with some proximal

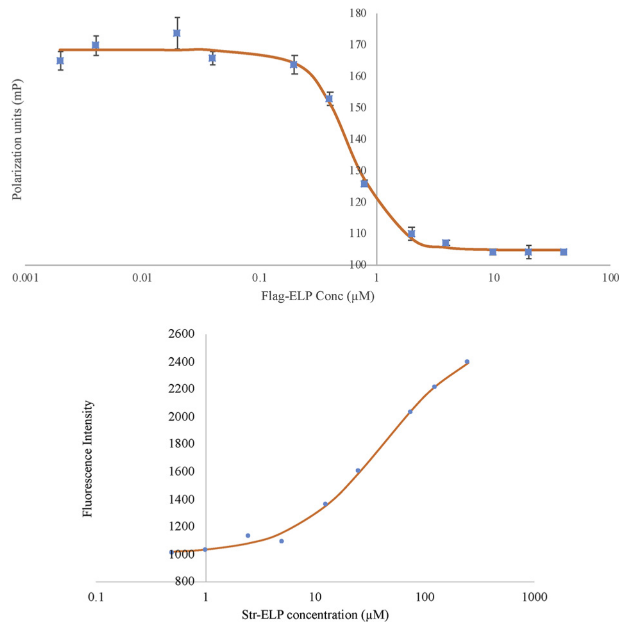


Fig. 3. Competitive fluorescence spectroscopy experiments to determine binding affinity of- (a) Flag-ELP to Anti-Flag Antibody M2 by competitive binding with labeled Flag peptide (n = 2) and (b) Str-ELP to streptavidin through competition with labelled streptavidin-binding peptide.

hydrophobic patches next to the binding region on the mAb. It is also possible that the Flag-ELP may have reduced entropic penalties for binding as compared to the free peptide ligand. These results with the Flag-ELP demonstrate that peptide-ELP fusions can exhibit strong binding to a given target.

To further examine if this binding of ELP peptide fusions was maintained for lower affinity, peptide-protein interaction systems, experiments were similarly carried out with the streptavidin-binding peptide (strep-pep) described above. Due to the many possible interactions of the strep-pep-ELP fusion with the tetrameric streptavidin, it was not possible to carry out a quantitative analysis of this interaction using a straightforward analysis using the BioEqs software. However, as can be seen in Fig. 3b, the data clearly indicated that the strep-pep-ELP construct was successfully able to displace the labeled peptide, thereby resulting in an increased fluorescence signal. Further confirmation of this interaction is obtained from the baseline approaching values (free labeled peptide has an intensity of 2600 IUs) of fluorescence intensities at the highest concentration of strep-pep-ELP. These results indicate that binding of strep-pep-ELP fusions retain their affinity for streptavidin.

In contrast, for the SMWRTYI-hGH system, little to no binding (data not shown) between hGH and the SMWRTYI-ELP construct was observed.

Table 2
Percentage capture and overall recovery for the purification of target proteins from crude mixtures using corresponding peptide-ELP fusions. Molar ratios between peptide-ELP and target protein have also been indicated.

System	Ratio of Peptide-ELP: Target	Capture (%)	Overall Recovery (%)
Flag-Antibody M2	8:1	98	95
Streptavidin (single peptide)	> 15:1	48	40
Streptavidin (di-peptide)	15:1	97	85
Human growth hormone	> 20:1	5	0

3.4. Evaluation of protein pulldown, recovery and purification by peptide-ELP fusions

After evaluating the binding of peptide-ELP affinity reagents, we studied their ability to purify target proteins via affinity precipitation. As indicated in Fig. 1, this separation process involves several steps. In this section, we evaluate the efficacy of peptide-ELP fusion based purification processes using two performance metrics: 1) capture efficiency – the ability to precipitate a complex of the target protein and the peptide-ELP fusion out of solution and 2) recovery – the release of the purified protein from the peptide-ELP fusion. Table 2 presents the percent of product captured in the first precipitation step as well as the overall recovery of the target protein after completion of the affinity precipitation process for the systems evaluated.

3.4.1. Purification of anti-Flag antibody M2

Since the antibody target possessed two homogeneous binding sites, one on either CDR region of the Anti-Flag M2, there would be a 2:1 molar ratio (stoichiometry) of binding to the antibody with complete complexation. Initial experiments were carried out with the pure antibody and the Flag-ELP to study the impact of stoichiometry on the capture efficiency. The results indicated that the Flag-ELP was successful in capturing greater than 80 % of the antibody at a 2:1 stoichiometry of peptide-ELP to Anti-Flag M2 antibody. Complete capture was achieved at higher ratios (4:1 and 8:1) (data not shown). Further, subsequent dissolution of the precipitate and elution at pH 3.0 was shown to recover majority of the antibody after removal of the ELP-Flag reagent in the second precipitation step. In addition, a 'blank' ELP construct without the peptide (ELP-only) was used as a control to ensure that protein capture was due exclusively to interactions with the affinity peptide.

After establishing pure component feasibility, the purification of Anti-Flag M2 mAb from complex mixtures was evaluated. The antibody was spiked into a crude *E. coli* lysate. Experiments were carried out in the *E. coli* system using stoichiometric ratios of 4:1 and 8:1 along with an intermediate precipitate wash and the results indicated that a ratio of 8:1 was required to achieve complete recovery of the antibody from this lysate. Fig. 4 presents the SDS PAGE gel analysis of various stages of this separation process. Lane 3 presents the initial feed solution consisting of the mAb spiked into the E coli lysate. The supernatant after the initial precipitation (capture step) is shown in lane 4. As can be seen, negligible amounts of the antibody were present, indicating efficient capture of antibody by the Flag-ELP in the presence of these impurities. On the other hand, 'supernatant 2′ (lane 5) consisted of essentially pure antibody. This indicates that the final supernatant after the dissolution and second precipitation step was successful at achieving high purity of the antibody from this complex mixture.

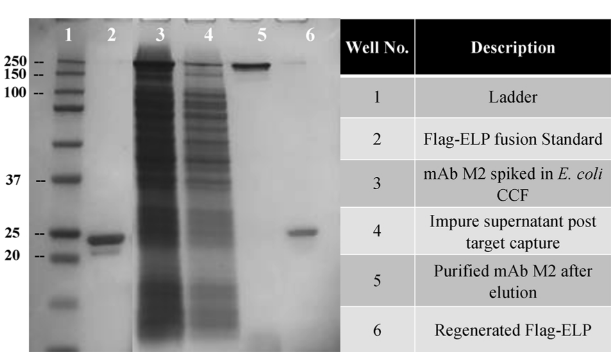


Fig. 4. Non-reducing SDS PAGE gel showing the purification of Anti-Flag antibody M2 (Lane 5) from a highly impure E. coli lysate (Lane 3) by Flag-ELP (Lane 2).

These results confirm that ELP-affinity peptide fusions with high affinity binders can indeed be successfully employed for affinity precipitation. To further test the limits of affinity desirable for the successful operation of this purification system, lower affinity peptides were evaluated with their corresponding target binding proteins. Accordingly, the affinity peptide for streptavidin discussed above (strep-pep) was examined as an ELP fusion for its utility to purify the target protein.

3.4.2. Purification of streptavidin

Streptavidin is a homo-tetrameric protein whose monomer has been shown to bind to the peptide DVEAWLDERVPLVET (Lamla and Erdmann, 2004). As discussed above, the free solution affinity of this peptide to streptavidin was weaker than the Flag peptide both as individual peptides and as ELP-fusions. Since streptavidin has 4 identical subunits, the stoichiometry of binding of strep-pep-ELP to streptavidin would be 4:1 at complete complexation, assuming the binding of one peptide to each monomer of the streptavidin. Affinity precipitation experiments were carried out with pure streptavidin in PBS with different stoichiometries (1:1, 2:1, 4:1, 8:1 and 10:1). The 'blank' ELP construct without the peptide (ELP-only) was used as a control to ensure that protein capture was due exclusively to interactions with the affinity peptide. Reversed-phase UPLC was employed to quantify streptavidin recovery and the results indicated that the ELP-only control did not result in any measurable capture of streptavidin. On the other hand, the strep-pep-ELP resulted in efficient capture of streptavidin, particularly at higher stoichiometries (Fig. 5).

Once the pure component capture of streptavidin was established,

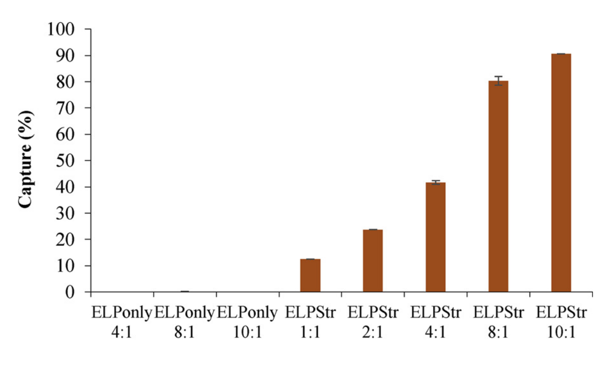
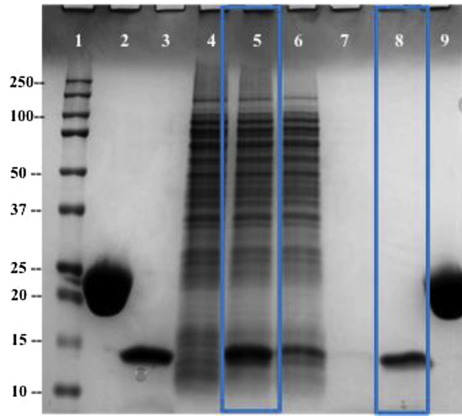


Fig. 5. Capture of streptavidin by ELP-only and ELPStr (strep-pep-ELP) at different molar binding ratios of ELP to streptavidin. While ELP-only captures negligible amounts of streptavidin, the ELPStr construct is successful in capturing streptavidin. The extent of capture, which increases with an increase in molar binding ratio, is found to saturate at higher molar ratios.



Well No.	Description
1	Ladder
2	Peptide-ELP fusion Standard
3	Pure Streptavidin Standard
4	Null E. coli cell lysate Standard
5	Streptavidin in cell lysate
6	Impure Supernatant post-capture
7	Wash step (x3)
8	Purified Streptavidin
9	Regenerated Peptide-ELP

Fig. 6. Non-reducing SDS PAGE gel showing the purification of streptavidin from *E. coli* cell lysate using strep-pep-ELP. Incomplete Streptavidin pulldown indicated by presence of unbound streptavidin in lane 6. Lane 8 shows recovery of pure streptavidin at levels comparable to the standard shown in lane 3.

we evaluated the ability of the strep-pep-ELP to selectively purify streptavidin from crude mixtures. Streptavidin was spiked into an *E. coli* lysate to a final concentration of 20 μM. As part of this evaluation, the elution behavior was examined at various pH and it was established that citrate buffer at pH 5.0 was effective in eluting streptavidin from the strep-pep-ELP. The entire process of binding, precipitation, washing and elution was then carried out with the crude mixture using a 10:1 ratio of peptide-ELP to streptavidin and SDS PAGE analysis of various steps in the purification are presented in Fig. 6. As can be seen in the gel, this process was successful at obtaining pure streptavidin (lane 8) which was also confirmed by RP-UPLC analysis. However, in contrast to the experiments with the pure streptavidin (Fig. 5), the capture efficiency with the crude feed was not as effective when using the 10:1 stoichiometry. This can be seen in lane 6 where there is still a streptavidin band in the post-capture supernatant. Efforts to increase the capture efficiency with the crude mixture using higher stoichiometries of peptide-ELP to streptavidin were not successful, with a maximum product recovery of 40 % being obtained. The inability to capture and recover higher amounts of streptavidin from the crude mixture could be due to several factors such as lower affinity (as compared to the Flag-ELP example) and/or the impact of competitive binding of impurities in the feed. Several strategies were examined to improve the recovery of streptavidin as described in the next section.

3.4.3. Improving streptavidin recovery

The peptide-ELP fusion employed above included a five amino acid long linker (G-S-G-S-G) between the peptide and the ELP sequence. To further increase solvent accessibility of the peptide, the length of this linker was increased to ten amino acid residues (G-S-G-S-G-S-G-S-G-S). However, this increase in linker length did not have a measurable effect on the streptavidin recovery from the crude mixture, indicating that solvent exposure was not the primary reason for the lower recovery.

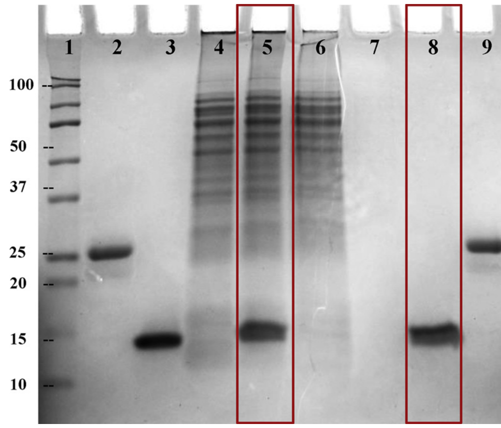
Since streptavidin exists as a homo-tetramer, there was potentially an opportunity to exploit multivalent interactions. We hypothesized that the use of bivalent constructs could potentially increase the apparent affinity of interaction due to avidity effects. Accordingly, a construct was created with two copies of the streptavidin-binding peptide in series to achieve a ‘peptide-linker-peptide-linker-ELP’ or (strep-pep)$_2$-ELP configuration. The linkers in this construct consisted of nine amino acids with alternating Gly-Ser residues. Experiments were carried out with binding stoichiometries of 12:1, 15:1 and 20:1. Interestingly, these results indicated that for the two higher stoichiometries the dipeptide-ELP fusion construct was able to completely capture and precipitate the streptavidin from the complex mixture. This can be seen in the gel presented in Fig. 7, where the streptavidin in *E. coli* lysate (lane 5) is absent from the supernatant post capture presented in lane 6. For the elution step, it was observed that an elution pH of 4.8 was more effective in completely dissociating the complex, likely due to enhanced affinity. The final product (shown in lane 8) from this process was obtained at 97 % purity as determined by RP-UPLC. Further, the overall recovery of this process was found to be > 85 % as determined by RP-UPLC analysis (Table 2). These results demonstrate that the re-designed dipeptide-ELP construct with optimized elution conditions was successful in purifying the tetrameric streptavidin with high yield and purity.

The improved capture efficiency of the dipeptide-ELP as compared to the single peptide-ELP construct could be due to a potential bivalent interaction between the two peptide units on the ELP and the tetrameric streptavidin which could increase the overall apparent affinity of the interaction. This is also supported by the requirement of a lower elution pH (4.8) required for the dipeptide construct. These results are in agreement with prior affinity precipitation studies carried out with an ELP-ZZ construct (2 Z domain moieties from protein A fused in series at the C-terminal end of ELP) which indicated increased recoveries of monoclonal antibodies as compared to ELP-Z constructs (Madan et al., 2013). It is important to note that despite the improved performance of the dipeptide-ELP construct, a high stoichiometry (15:1) was still required for complete capture of the streptavidin from the complex feed. In contrast, an 8:1 ratio of the higher affinity single Flag-peptide-ELP construct was sufficient for complete capture of the antibody. These results further confirm that peptide-protein affinity plays an important role in the overall purification performance that can be achieved via peptide-ELP fusion-mediated affinity precipitation.

3.4.4. Evaluation of hGH purification

In order to further study the impact of peptide-protein affinity on the ELP-driven affinity precipitation process we examined a lower affinity peptide. As indicated earlier, the affinity of the peptide SMWRTYI – human growth hormone (hGH) interaction was 32 μM as compared to 7.54 u M and 99.4 nM for the streptavidin and Flag systems, respectively. A SMWRTYI-ELP fusion separated by a G-S-G-S-G-S linker was produced and evaluated for its efficacy in capturing hGH from solution. A range of operating parameters (binding stoichiometry, pH) and linker chemistries (flexible, rigid and charged) were evaluated to find a condition that would enable binding of SMWRTYI-ELP with hGH. However, none of the conditions evaluated could successfully capture and precipitate hGH along with the peptide-ELP construct. In contrast, previous results with this peptide used as a chromatographic ligand were able to recover the hGH.

It is important to note that while peptide-target affinity is important, other factors such as fusion with a hydrophobic ELP, inherent hydrophobicity of the peptide and appropriate linkers between the peptide and ELP could also affect the success of the process. It is likely that the affinity between the peptide and hGH was not high enough to



Well No.	Description
1	Ladder
2	Di-peptide-ELP fusion Standard
3	Pure Streptavidin Standard
4	Null E. coli cell lysate Standard
5	Streptavidin in cell lysate
6	Impure supernatant post-capture
7	Wash step (x3)
8	Purified Streptavidin
9	Regenerated Di-peptide-ELP

Fig. 7. Non-reducing SDS PAGE gel showing the purification of streptavidin from *E. coli* cell lysate using a dipeptide-ELP ((strep-pep)$_2$-ELP) fusion. Absence of streptavidin in lane 6 confirms complete pulldown. 97 % pure streptavidin was obtained with an 85 % recovery in lane 8 (comparable to standard in lane 3).

keep the hGH bound to the SMWRTYI-ELP conjugate during the first precipitation step. Further, the hydrophobicity of the peptides and ELP were examined. GRAVY hydropathy scores for the peptides and ELP are presented in SI Table 1. It can be observed that the SMWRTYI and streptavidin-binding peptide are more hydrophobic than the Flag peptide. SMWRTYI contained fewer charged/hydrophilic residues (14 %) as compared to the streptavidin-binding peptide and Flag which had 40 and 88 %, respectively. The ELP sequence was also observed to possess a significantly high hydropathy score (0.878) with a frequent occurrence of valines. Thus, in addition to the lower affinity, we hypothesize that the relatively higher hydrophobic content and smaller size of the SMWRTYI peptide could have led to partial peptide-ELP association thereby reducing the solvent accessibility of the peptide required for interacting with hGH.

4. Conclusions

The results in this paper demonstrate that affinity peptide - smart biopolymer fusions can indeed be employed for the purification of proteins via affinity precipitation. The affinity of the peptide-protein interaction is shown to be important. For high affinity systems (e.g. FLAG), these processes are very effective resulting in both high purity and yield with minimal process development required. On the other hand, for intermediate affinity systems (e.g. streptavidin-binding peptide), it was found that avidity interactions can be useful in improving performance. This multivalency-based affinity precipitation strategy may also be useful for the purification of other therapeutic multimeric or multi-domain proteins as well as more complex biological systems. The results also demonstrated that for lower affinity systems (e.g. hGH binding peptide) these processes can be problematic, indicating that sufficiently high affinity is required for these precipitation processes to be successful. However, due to potential complications of hydrophobic interactions between peptide and ELP as well as peptide solvent exposure, future studies will need to be carried out to more clearly establish the lower limits of affinity between the peptide and target for this particular affinity precipitation approach.

The ability to rapidly identify affinity peptides using a range of high throughput and screening techniques combined with the ease of producing these peptide-ELP constructs using standard recombinant approaches opens up significant opportunities for producing new affinity capture reagents for biological products. This may be particularly important for biological products without an established affinity capture platform which has proven so useful for mAbs using protein A. In addition, this approach may have potential for important new classes of biological products such as gene therapy agents. Finally, the ease of carrying out precipitation processes in a continuous fashion may be an additional opportunity for this new class of affinity peptide-smart biopolymer systems.

CRediT authorship contribution statement

Akshat Mullerpatan: Conceptualization, Methodology, Investigation, Validation, Formal analysis, Writing - original draft. **Divya Chandra:** Methodology, Investigation, Writing - review & editing. **Erin Kane:** Investigation, Validation. **Pankaj Karande:** Writing - review & editing, Supervision, Project administration, Funding acquisition. **Steven Cramer:** Writing - review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.jbiotec.2019.12.012.

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